

**The National University of Singapore
School of Computing**

CS4226 – Internet Architecture

Tutorial 2

Week 4: August 31 - September 4, 2026

NOTE:

- Please work through the tutorial problems before class and bring your solutions with you.
- If any part of a question is unclear, you are encouraged to ask on Piazza.

Q1. Suppose the arrivals of A2 buses follow a Poisson process with rate λ . What is the probability that two consecutive arrivals come within a time period of length t ?

Q2. Suppose the arrivals of A2 buses follow a Poisson process with rate λ . When you arrive at the bus stop, you are told that the last A2 bus has departed for s amount of time. What is the probability that your waiting time for the next A2 bus is longer than t ?

Q3. Suppose the arrivals of A2 buses and D1 buses follow Poisson processes with rate λ_{A2} and λ_{D1} . Suppose you want to go to the central library from SoC and can take either A2 or D1. Unfortunately, when you arrived at the bus stop, you see an A2 bus and a D1 bus have just departed. How long is your average waiting time for you to get on your bus?

Q4. Suppose we know that the probability that an M/M/1 system having i customers equals $\pi_i = \rho^i(1 - \rho)$. Derive the expected number of customers in the system. How about the expected number of customers in the queue?